

Varun Raveendra

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Salt Lake City, UT

Education

PhD, Robotics University of Utah	<i>Jan 2025 - Present</i>
Master's, Electrical and Computer Science University of Utah	<i>Aug 2022 - May 2024</i>
Bachelor's, Electronics and Communication Visvesvaraya University, NMIT	<i>Jun 2018 - Jul 2022</i>

Skills

Programming Languages: Python, C, C++, Embedded C, MATLAB.
Software Tools: Fusion 360, Webots, SolidWorks, Gazebo, Isaac Sim, Git, Docker
Software Frameworks: ROS2, PyTorch, TensorFlow, Serial Manipulators
Core Competencies: Robot Manipulators and Control, Motion Planning, Deep Learning, Machine Learning, Image Processing, Embedded Systems, Computer Networks, Computer Vision, Microcontrollers, Moveit2.
Other Skills: 3D Printing, Arduino, Raspberry Pi.

Publications

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- **Emergent Swarm Behavior Prediction using 1D-Convolution for Autonomous Closed-Loop Behavior Control** *ARMS 2025 Workshop (AAMAS 2025)*
Varun Raveendra, Connor Mattson, Daniel S. Brown
 - **Discovery and Deployment of Emergent Robot Swarm Behaviors via Representation Learning and Real2Sim2Real Transfer** *AAMAS 2025*
Connor Mattson, *Varun Raveendra*, Ricardo Vega, Cameron Nowzari, Daniel S. Drew, Daniel S. Brown
 - **Waste management in urban localities: an IoT and machine learning solution** *IJSAEM 2025*
V. Viswanatha, Rony Joseph Theckeveetil, *Varun Raveendra*, Sreeteja Tummala, K. M. Suhas

Work Experience

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- **Research Assistant (ARIA Lab)**, Khalert School of Computing, University of Utah *Jan 2025 – Present*
Part of a lab that researches on Reinforcement Learning, Human-Robot Interaction and Multi-Agent robots. Currently researching on a multidisciplinary team to design and research multi-agent robots at the University of Utah.
 - **Graduate Teaching Assistant - Motion Planning**, University of Utah *Jan 2025 – May 2025*
Sole TA responsible for managing all aspects of student support, including grading assignments, addressing lingering questions, and re-teaching key concepts to ensure comprehension. Acted as the primary point of contact for over 70 students, balancing instructional duties with academic evaluation.
 - **Research Engineer**, Khalert School of Computing, University of Utah *Aug 2024 – Dec 2024*
Leading a multidisciplinary team to design and research swarm robots at the University of Utah.
 - **Graduate Teaching Assistant - Circuits**, University of Utah *Aug 2023 – Dec 2023*
Integrated engaging teaching techniques that increased students' troubleshooting proficiency and understanding of essential concepts by 30%; attendance surged by 5% during lab sessions, reflecting increased student engagement.
 - **Embedded Systems Intern**, Tevatron Technologies *Jun 2021 – Aug 2021*
Designed and documented schematics for STM32F401RE boards, achieving a 20% increase in project workflow efficiency, allowing for quicker adaptations to design changes based on evolving application requirements.

Projects

- **Adaptive learning for Open Agent Systems - Winner of MOASEI 2025!** *April 2025 - May 2025*
Trained GNN-based RL agents capable of adapting to open environments and changes without relying on direct communication or a central controller. We focus on a wildfire scenario where robots with limited resources collaborate to extinguish randomly appearing and spreading fires.
- **Emergent Behaviour in Swarm Robots - HeRo+** *May 2024 - December 2024*
Initiated a pilot program building swarm robots and unified a comprehensive research methodology for analyzing swarm robot interactions, leading to the discovery of novel behaviors that accelerated collaborative tasks.
- **Localization and control of 3D printed autonomous robots** *Jan 2024 - May 2024*
Revised a real-time localization system that utilized camera data, improving localization accuracy by 40% in expansive environments, enabling precise navigation for swarm robotics applications.
- **Design of a micro robotic continuum gripper for medical applications** *Aug 2022 - Jan 2024*
Lead and created a groundbreaking micro robotic gripper prototype for minimally invasive surgeries, reducing average surgery time by 25 minutes and 10% reduction in recovery time for patients.
- **Motion Planning collision - Multi-Agent Path Plans (MAPF)** *Feb 2023 - Apr 2023*
Deployed a multi-agent path planning software that facilitated collision-free navigation for 50+ robotic agents, enhancing operational efficiency and reducing average travel time by 25%.
- **SLAM using Crazyflie** *Aug 2022 - Dec 2022*
Teamed and implemented the integration of Nav2 with ROS2 on the Crazyflie drone. Created and tested a comprehensive simulation environment to assess the drone's mapping capabilities.
- **Neural network aided robotic separator for domestic waste** *Mar 2022 - Aug 2022*
Spearheaded the team in creating and testing a machine learning algorithm that simplified the identification of recyclable materials, increasing the overall efficiency of waste management operations by 25% and enhancing sustainability practices.

Coursework

- Intro to Robot Kinematics and Dynamics
- Robot Manipulation and Control
- Digital Image Processing
- Computer Vision
- Motion Planning
- Robotics Millisystems
- Machine Learning
- Deep learning
- Advanced AI for Robotics
- Advanced Robot Algorithms